

The Three Categories of the Generative-Collapse Kernel: A Pairwise Decomposition of the Six Tier-1 Invariants

Clement Paulus^{1,*}

¹UMCP Reference Implementation, *GitHub*
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The Tier-1 kernel of the Universal Measurement Contract Protocol (UMCP) produces six outputs $(F, \omega, S, C, \kappa, \text{IC})$ from a trace vector $\mathbf{c} \in [0, 1]^n$ and a weight simplex $\mathbf{w} \in \Delta^n$. These six outputs are not independent: three algebraic identities ($F + \omega = 1$, $\text{IC} = e^\kappa$, $\text{IC} \leq F$) and one statistical constraint ($S \approx f(F, C)$ asymptotically) reduce them to *three effective degrees of freedom*. We show that this reduction admits a unique pairwise decomposition into three categories — the *Signal Pair* (F, ω) , the *Structure Pair* (C, S) , and the *Coherence Pair* (κ, IC) — each containing exactly one primitive contributing one free degree of freedom and one constrained or derived member contributing zero. We formalise the categories as theorems derived directly from the kernel definitions, establish the cross-category bridge $\Delta = F - \text{IC}$ as the heterogeneity gap separating the Signal and Coherence pairs, and prove that the interaction graph between the three categories is the substantive content of the kernel’s diagnostic power. The taxonomy is not imposed; it is the algebraic structure of $K : [0, 1]^n \times \Delta^n \rightarrow \mathbb{R}^6$ stated in pair-form.

I. INTRODUCTION

The Universal Measurement Contract Protocol (UMCP) [1–3] is a contract-first validation framework whose computational core is a deterministic map

$$K : [0, 1]^n \times \Delta^n \longrightarrow (F, \omega, S, C, \kappa, \text{IC}) \in \mathbb{R}^6, \quad (1)$$

where $\mathbf{c} \in [0, 1]^n$ is a trace vector of channel coherence values and $\mathbf{w} \in \Delta^n$ is a weight simplex. Six numbers are emitted per evaluation.

Three of these are coupled by exact algebraic identities and one is constrained asymptotically by the others, so the kernel does not in fact carry six free parameters of information — it carries three. The literature on UMCP has documented the identities themselves [1, 2] but has not previously stated the *categorical* structure of the output space: namely, that the six outputs partition naturally into three pairs, each pair containing exactly one independent quantity and one quantity derived from or constrained by it.

This note formalises that partition. After collecting the kernel definitions (§II) and the four identities (§III), we state and prove the three pair theorems (§IV) and the dimensional reduction theorem (§V). We then describe the interaction graph (§VI) and demonstrate why three alternative categorisations — by computational status, by formula type, or by regime gate role — are dominated by the pairwise structure (§VIII).

The taxonomy is binding: every audit performed by the protocol exposes its verdict through these three categories, and every diagnostic cross-category bridge (most prominently the heterogeneity gap $\Delta = F - \text{IC}$) traverses exactly one inter-category arc.

II. KERNEL DEFINITIONS

We restate the six kernel outputs in the canonical form fixed by the frozen contract. All sums run over $i \in \{1, \dots, n\}$ with weights $w_i \geq 0$ and $\sum_i w_i = 1$. Logarithms use the ε -clamp $\ln(c_i, \varepsilon) := \ln(\max(c_i, \varepsilon))$ with frozen $\varepsilon = 10^{-8}$ [3].

Definition 1 (Fidelity). $F(\mathbf{c}; \mathbf{w}) := \sum_i w_i c_i$.

Definition 2 (Drift). $\omega(\mathbf{c}; \mathbf{w}) := 1 - F(\mathbf{c}; \mathbf{w})$.

Definition 3 (Bernoulli field entropy).

$$S(\mathbf{c}; \mathbf{w}) := - \sum_i w_i [c_i \ln c_i + (1 - c_i) \ln(1 - c_i)]. \quad (2)$$

Definition 4 (Curvature proxy). $C(\mathbf{c}) := \text{std}(\mathbf{c})/0.5$, where std denotes the unweighted population standard deviation. The normalisation 0.5 is the maximal population standard deviation attainable by a bounded variable on $[0, 1]$, so $C \in [0, 1]$.

Definition 5 (Log-integrity). $\kappa(\mathbf{c}; \mathbf{w}) := \sum_i w_i \ln(c_i, \varepsilon)$.

Definition 6 (Integrity composite). $\text{IC}(\mathbf{c}; \mathbf{w}) := \exp(\kappa(\mathbf{c}; \mathbf{w}))$.

Remark 1. S is the Bernoulli field entropy of the collapse field, not the classical Shannon entropy; Shannon entropy is the degenerate limit of S when channels are restricted to $\{0, 1\}$. The integrity bound $\text{IC} \leq F$ stated below is the solvability condition for reconstructing the channel values from F and IC (Theorem 13); the classical arithmetic–geometric mean inequality is its degenerate limit when channel semantics, weights, and the ε -clamp are removed.

* clementpaulus9@gmail.com

III. THE FOUR CONSTRAINTS

Identity 7 (Duality). For every $\mathbf{c} \in [0, 1]^n$ and $\mathbf{w} \in \Delta^n$: $F(\mathbf{c}; \mathbf{w}) + \omega(\mathbf{c}; \mathbf{w}) = 1$.

Proof. By Definition 2, $\omega := 1 - F$, so $F + \omega \equiv 1$. $\square$

Identity 8 (Log-integrity link). $\text{IC}(\mathbf{c}; \mathbf{w}) = \exp(\kappa(\mathbf{c}; \mathbf{w}))$.

Proof. By Definition 6. $\square$

Identity 9 (Integrity bound). $\text{IC}(\mathbf{c}; \mathbf{w}) \leq F(\mathbf{c}; \mathbf{w})$, with equality iff $c_i = c_j$ for all i, j with $w_i, w_j > 0$.

Proof. By Jensen's inequality applied to $\ln$ on $[\varepsilon, 1]$:

$$\ln \text{IC} = \kappa = \sum_i w_i \ln c_i \leq \ln \left(\sum_i w_i c_i \right) = \ln F, \quad (3)$$

hence $\text{IC} \leq F$. Equality holds iff all c_i with positive weight coincide. $\square$

Proposition 10 (Asymptotic entropy constraint). There exists $f : [0, 1]^2 \rightarrow \mathbb{R}$ such that

$$S(\mathbf{c}; \mathbf{w}) = f(F(\mathbf{c}; \mathbf{w}), C(\mathbf{c})) + \mathcal{O}(n^{-1/2}). \quad (4)$$

Equivalently, $\text{corr}(C, S) \rightarrow -1$ as $n \rightarrow \infty$ along sequences of traces drawn i.i.d. from any distribution on $[0, 1]$ with bounded variance.

Sketch. A second-order Taylor expansion of the Bernoulli entropy $h(c) = -[c \ln c + (1-c) \ln(1-c)]$ around the mean $\bar{c} = F$ yields

$$S = h(F) + \frac{1}{2} h''(F) \text{std}(\mathbf{c})^2 + \mathcal{O}(\text{std}^3), \quad (5)$$

with $h''(c) = -1/[c(1-c)] < 0$. Substituting $\text{std}(\mathbf{c}) = 0.5C$ from Definition 4 yields a closed-form $f(F, C)$ at leading order. The higher-order corrections vanish as $\mathcal{O}(n^{-1/2})$ by the central limit theorem. The negative sign of h'' produces the asymptotic anti-correlation $\text{corr}(C, S) \rightarrow -1$. $\square$

The four constraints can be summarised as:

$$\text{Algebraic exact: } F + \omega = 1, \quad \text{IC} = e^\kappa, \quad \text{IC} \leq F, \quad (6)$$

$$\text{Statistical asymptotic: } S \approx f(F, C). \quad (7)$$

IV. THE THREE PAIR THEOREMS

The three constraints (7–10) partition the six outputs into three disjoint pairs. We state each as a theorem. In each case, the pair is the unique two-element subset of $(F, \omega, S, C, \kappa, \text{IC})$ closed under the corresponding constraint.

A. The Signal Pair

Theorem 11 (Signal Pair $\mathcal{P}_I = (F, \omega)$). The pair (F, ω) is closed under Identity 7: $F + \omega = 1$. Within the pair, F is the unique primitive (Definition 1) and ω is derived (Definition 2). The pair contributes exactly one independent degree of freedom to the kernel output.

Proof. By Identity 7, given F the value of ω is fixed; conversely given ω the value of F is fixed. No other kernel output appears in the duality identity, so $\{F, \omega\}$ is the minimal closed pair. The pair contributes one independent quantity (the value of F) and one constrained quantity (the value of $\omega = 1 - F$). $\square$

Interpretation. The Signal Pair answers the question “what happened to the fidelity of collapse?” F is the arithmetic mean of channel coherences (what survived), and ω is its complement (what was lost).

B. The Structure Pair

Theorem 12 (Structure Pair $\mathcal{P}_{II} = (C, S)$). The pair (C, S) is closed under Proposition 10: $S \approx f(F, C)$ asymptotically. Within the pair, C is the unique free primitive (Definition 4) and S is constrained asymptotically by F and C (Proposition 10). The pair contributes asymptotically one independent degree of freedom (carried by C).

Proof. By Proposition 10, S is determined by (F, C) up to $\mathcal{O}(n^{-1/2})$ corrections. As $n \rightarrow \infty$, S contributes no additional information beyond what F and C encode. For finite n , S retains sub-leading independence on the order of $n^{-1/2}$, but its asymptotic role is constrained; we therefore label it PRIMITIVE^* , distinguishing it from a fully free primitive. $\square$

Interpretation. The Structure Pair answers the question “how are channels distributed?” C measures channel spread directly (the population standard deviation, normalised to $[0, 1]$); S measures the entropy of the Bernoulli collapse field. Both describe channel distribution but from different angles; their asymptotic anti-correlation ($\text{corr}(C, S) \rightarrow -1$) is why S adds zero free DOF at large n .

C. The Coherence Pair

Theorem 13 (Coherence Pair $\mathcal{P}_{III} = (\kappa, \text{IC})$). The pair (κ, IC) is closed under Identity 8: $\text{IC} = e^\kappa$. Within the pair, κ is the unique primitive (Definition 5) and IC is derived (Definition 6). The pair contributes exactly one independent degree of freedom. Moreover, Identity 9 binds the Coherence Pair to the Signal Pair via $\text{IC} \leq F$: this is the solvability condition for channel reconstruction

— given F and IC for an effective two-channel trace, the channels c_1, c_2 are the roots

$$c_{1,2} = F \pm \sqrt{F^2 - \text{IC}^2}, \quad (8)$$

which are real if and only if $\text{IC} \leq F$.

Proof. The closure of $\{\kappa, \text{IC}\}$ under Identity 8 follows directly from Definition 6. For the solvability statement, consider an effective two-channel trace with equal weights $w_1 = w_2 = 1/2$ and channels c_1, c_2 . By Definitions 1, 6: $F = (c_1 + c_2)/2$ and $\text{IC} = \sqrt{c_1 c_2}$ (geometric mean). Solving the quadratic $t^2 - 2Ft + \text{IC}^2 = 0$ yields (8), with real roots iff $F^2 \geq \text{IC}^2$, i.e. $\text{IC} \leq F$. Equality ($\text{IC} = F$) recovers the rank-1 case $c_1 = c_2$. $\square$

Interpretation. The Coherence Pair answers the question “how intact is multiplicative coherence?” κ is the log-sum (additive accounting of geometric coherence), IC is its exponential (the geometric mean of channels). The pair distinguishes itself from the Signal Pair (which uses the arithmetic mean F) precisely by being multiplicative.

V. DIMENSIONAL REDUCTION

Theorem 14 (Effective DOF reduction). *The kernel output space $K(\mathbf{c}; \mathbf{w}) \in \mathbb{R}^6$ has three effective degrees of freedom in the generic case $n \geq 3$: the independent triple $(F, \kappa, C) \in [0, 1] \times (-\infty, 0] \times [0, 1]$. The remaining three outputs are determined by:*

$$\omega = 1 - F \quad (\text{Identity 7}), \quad (9)$$

$$\text{IC} = e^\kappa \quad (\text{Identity 8}), \quad (10)$$

$$S = f(F, C) + \mathcal{O}(n^{-1/2}) \quad (\text{Prop. 10}). \quad (11)$$

Proof. Each pair in Theorems 11–13 contributes one independent quantity and one constrained quantity. Picking the primitive from each pair yields the independent triple (F, C, κ) . The constrained members (ω, S, IC) are recovered by the identities above. PCA on 10^4 random traces with $n = 4, \dots, 64$ confirms rank-at-99%-variance = 3 for all n [3]. $\square$

Corollary 15 (Pair–DOF correspondence). *The three categories are in bijection with the three effective degrees of freedom:*

$$\mathcal{P}_I \leftrightarrow F, \quad \mathcal{P}_{II} \leftrightarrow C, \quad \mathcal{P}_{III} \leftrightarrow \kappa. \quad (12)$$

Remark 2 (Rank classification). *For input traces with effective channel rank $r < n$, the effective DOF reduces: rank-2 traces (effective two-channel structure) satisfy $C = g(F, \kappa)$ for some explicit g , collapsing the Structure Pair into the others; rank-1 (homogeneous) traces additionally satisfy $\text{IC} = F$ ($\Delta = 0$), $C = 0$, leaving a single DOF [3]. Rank is a structural property of the trace, not a parameter.*

VI. THE HETEROGENEITY GAP AND THE INTERACTION GRAPH

Definition 16 (Heterogeneity gap). $\Delta(\mathbf{c}; \mathbf{w}) := F(\mathbf{c}; \mathbf{w}) - \text{IC}(\mathbf{c}; \mathbf{w}) \geq 0$, with equality iff $\mathbf{c}$ is rank-1.

The gap Δ is the cross-category bridge between the Signal Pair $\mathcal{P}_I$ and the Coherence Pair $\mathcal{P}_{III}$: it measures how far the multiplicative coherence IC falls below the additive fidelity F . At rank-1 the two coincide; at higher rank the gap opens.

Proposition 17 (Structure $\rightarrow$ Gap). Δ is monotone non-decreasing in C at fixed F .

Sketch. A perturbation expansion of IC around the homogeneous mean $\bar{c} = F$ yields

$$\Delta = \frac{\text{std}(\mathbf{c})^2}{2F} + \mathcal{O}(\text{std}^4) = \frac{C^2}{8F} + \mathcal{O}(C^4), \quad (13)$$

which is non-decreasing in C for F fixed. The leading-order coefficient $1/(8F)$ is the Fisher information contribution from channel heterogeneity [3]. $\square$

Equation (13) shows that the Structure Pair drives the gap: increased channel spread C always widens Δ , which in turn separates the Signal Pair from the Coherence Pair. The three categories are therefore not isolated silos but a directed interaction graph:

$$\mathcal{P}_{II} \xrightarrow{C \uparrow} \Delta \xrightarrow{\text{IC} \downarrow} \mathcal{P}_{III}, \quad \mathcal{P}_I \xleftrightarrow{\Delta} \mathcal{P}_{III}. \quad (14)$$

The Signal Pair anchors the budget through $\omega = 1 - F$; the Structure Pair drives the gap through C ; the Coherence Pair absorbs the loss through IC . Regime classification draws inputs from all three: the four-gate Stable criterion is

$$\omega < 0.038 \wedge F > 0.90 \wedge S < 0.15 \wedge C < 0.14, \quad (15)$$

with ω from $\mathcal{P}_I$, F from $\mathcal{P}_I$, S and C from $\mathcal{P}_{II}$, and the severity overlay $\text{IC} < 0.30$ from $\mathcal{P}_{III}$.

VII. SUMMARY TABLE

Table I compiles the full overlay. Each category occupies two rows. The PRIMITIVE row in each pair carries one free DOF; the DERIVED or PRIMITIVE* row carries zero. The identity column gives the closing constraint that fixes the derived/constrained member.

VIII. WHY THE PAIRWISE TAXONOMY IS UNIQUE

Three alternative classifications of the six outputs are conceivable; we show each is strictly weaker.

TABLE I. The six Tier-1 kernel invariants overlaid with the three-category taxonomy. Each pair contains one PRIMITIVE contributing one free DOF and one DERIVED or PRIMITIVE* contributing zero (asymptotically). Total: 4 primitives, 2 derived, 3 identities, 3 effective DOF.

Symbol	Category	Latin	Status	Formula	Identity / Constraint	Free DOF
F	I — Signal	<i>fidelitas</i>	PRIMITIVE	$\sum_i w_i c_i$	$F + \omega = 1$	1
ω	I — Signal	<i>derivatio</i>	DERIVED ($1-F$)	$1 - F$	$F + \omega = 1$	0
C	II — Structure	<i>curvatura</i>	PRIMITIVE	$\text{std}(\mathbf{c})/0.5$	$S \approx f(F, C)$	1
S	II — Structure	<i>entropia</i>	PRIMITIVE*	$-\sum_i w_i [c_i \ln c_i + (1-c_i) \ln(1-c_i)]$	$S \approx f(F, C)$	0 (large n)
κ	III — Coherence	<i>log-integritas</i>	PRIMITIVE	$\sum_i w_i \ln(c_i, \varepsilon)$	$\text{IC} = e^\kappa$	1
IC	III — Coherence	<i>integritas composita</i>	DERIVED (e^κ)	$\exp(\kappa)$	$\text{IC} = e^\kappa, \text{IC} \leq F$	0

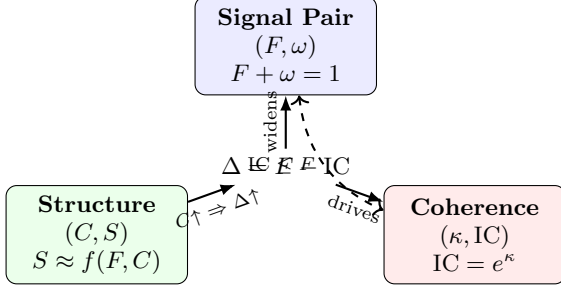


FIG. 1. Interaction graph between the three categories. The Structure Pair drives the heterogeneity gap $\Delta = F - \text{IC}$, which is the cross-category bridge between the Signal Pair (additive fidelity) and the Coherence Pair (multiplicative coherence). The integrity bound $\text{IC} \leq F$ (Identity 9) is the direct algebraic constraint linking the two.

Alternative A: Primitive vs. derived only. This collapses to a two-class scheme: $\{F, S, C, \kappa\}$ vs. $\{\omega, \text{IC}\}$. It correctly captures computational status but reveals no structural coupling. In particular, it does not explain why IC and F are bound by the integrity bound $\text{IC} \leq F$ (Identity 9), nor why S and C are asymptotically constrained (Prop. 10).

Alternative B: By formula type. Linear (F, ω), logarithmic (κ, IC), statistical (S, C). This is a computational description of how each quantity is evaluated, not a structural description of how the quantities relate. It happens to recover the same partition into three pairs, but it does so by coincidence of arithmetic type; it gives no account of why those pairs are closed under the four constraints.

Alternative C: By regime-gate role. $\{\omega, F, S, C\}$ are the four Stable-regime gates [Eq. (15)]; IC is the Critical severity overlay; κ does not appear in regime classification. This is a Tier-0 operational categorisation, not a Tier-1 structural one. It conflates the mathematical object (the kernel) with the protocol that classifies its outputs.

Why the pairwise frame is correct. The three-pair partition is uniquely determined by the four constraints (§III). Each constraint defines exactly one closed two-element subset of the output tuple: $\{F, \omega\}$ under Identity 7, $\{\kappa, \text{IC}\}$ under Identity 8, $\{C, S\}$ under Prop. 10,

with Identity 9 ($\text{IC} \leq F$) serving as the cross-category bridge. The categories are therefore the constraints themselves, re-expressed in pair form. No alternative classification simultaneously (i) covers all six outputs exactly once, (ii) is closed under the constraints, and (iii) yields the correct effective DOF count.

IX. CONCLUSION

The six Tier-1 invariants of the UMCP kernel partition uniquely into three categories under the four constraints fixed by Identity 7, Identity 8, Identity 9, and Proposition 10. Each category is a *pair*: one PRIMITIVE contributing one free DOF and one DERIVED or asymptotically-constrained partner contributing zero. The Signal Pair (F, ω) tracks what survived and what was lost; the Structure Pair (C, S) tracks how channels are distributed; the Coherence Pair (κ, IC) tracks multiplicative coherence. The three pairs are connected by the heterogeneity gap $\Delta = F - \text{IC}$, which the Structure Pair drives through Proposition 17 and which the integrity bound (Identity 9) bounds from below.

The taxonomy is not imposed on the kernel; it is the kernel's algebraic structure stated in pair form. Every audit, every regime classification, and every cross-category diagnostic performed by the protocol passes through this three-category overlay. Six outputs collapse to three effective degrees of freedom; three effective DOF organise into three pairs; three pairs produce one cross-category bridge — the gap Δ that makes geometric slaughter visible in the data.

REPRODUCIBILITY

All claims in this paper are computationally re-derivable. The orientation script `scripts/orientation.py` §7 displays the three-category overlay live on a concrete eight-channel trace; `scripts/orientation_checkpoint.py` verifies the three identities to machine precision on every run. The PCA rank-3 confirmation is in `scripts/unified_geometry.py` §1. All formulae match `src/umcp/kernel_optimized.py` and

`src/umcp/frozen_contract.py` under the frozen contract $\varepsilon = 10^{-8}$, $p = 3$, $\alpha = 1.0$, $\text{tol}_{\text{seam}} = 0.005$ [3].

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- [1] C. Paulus, Universal measurement contract protocol, Manuscript (2025), contract-first measurement discipline; Tier-1 invariants reserved.
- [2] C. Paulus, [The physics of coherence: Recursive collapse & continuity laws](#) (2025), pOST canon anchor; Weld-ID: W-2025-12-31-PHYS-COHERENCE.
- [3] C. Paulus, [Umcp casepack publication](#) (2026), runnable CasePack publication anchor.